Dataset Identification:-

- Dataset: Amazon Sampled Product Graph
- Source File: amazon_sampled_graph.csv
- 1,000 Total Nodes
- 2,178 Total Edges

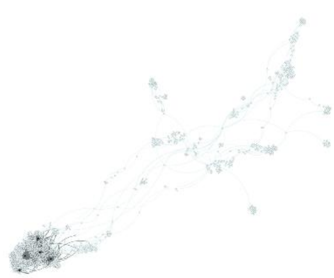
Key Network Influencers:-

- Centrality analysis was performed to identify highly connected hubs and influential bridges within the network.

Rank	Top 5 by Degree (Popularity)	Top 5 by Betweenness (Influence)
1	Node 458358	Node 458358
2	Node 175102	Node 161797
3	Node 38365	Node 215057
4	Node 272396	Node 380964
5	Node 399385	Node 516125

Visualizations:-

Id	Label	Interval	Degree	Modularity Class	Eccentricity	Closeness Centrality	Harmonic Closeness Centrality	Betweenness Centrality	PageRank
458358			324	1	7.0	0.309288	0.519214	235624.81744	0.069371
38365			151	3	7.0	0.290914	0.427306	101842.664473	0.033413
175102			92	3	7.0	0.277731	0.384246	48184.635587	0.019378
498702			46	3	8.0	0.250753	0.360916	6770.201922	0.008818
37981			46	1	8.0	0.239052	0.331114	903.320251	0.00827
521221			41	0	7.0	0.244374	0.293222	27510.996024	0.009512
222939			32	2	6.0	0.187747	0.217501	24306.017944	0.006593
306602			27	5	7.0	0.16412	0.197092	17209.9322	0.004934
30049			26	6	7.0	0.2002	0.223085	23618.066667	0.007361
219358			21	11	5.0	0.266045	0.290641	75759.664494	0.004693
68409			20	4	6.0	0.187747	0.210127	13258.991114	0.00478
138688			19	12	6.0	0.18652	0.206607	17321.570477	0.004016



Community Detection & Interpretation of Results:-

- **Insight:** Each color cluster likely represents a specific product category (e.g., Books, Electronics, Gardening).
- **Observations:** The visualization shows high "within-community" density, meaning customers who buy one item in a category are extremely likely to buy several more from the same group.

Filtering & Highlighting Analysis:-

- **Hub Identification:** The Top 5 degree nodes were scaled up and **colored black**. These nodes represent the "Gravity Wells" of the network—if these items are out of stock, it likely impacts the sales of hundreds of connected items.
- **Bridge Analysis:** High Betweenness Centrality nodes were highlighted to identify products that connect two different communities. These are "cross-over" products that facilitate discovery across different departments.

Key Findings & Learning:-

1. **Network Resilience:** The Amazon graph is a "Scale-Free" network. It is highly dependent on a few massive hubs.
2. **Recommendation Engine Logic:** This analysis provides the backbone for "Customers who bought this also bought..." features. High degree nodes are the safest recommendations, while high betweenness nodes are the best for cross-selling.
3. **Technical Proficiency:** I successfully moved from raw co-purchase data to a modular visualization, learning how to use Modularity to automatically categorize products without needing manual labels.